Static and kinetic friction

LEARNING TARGET I can distinguish adjustable static friction from kinetic friction.

A · KEY VOCABULARY

Static f_s	Adjusts to prevent slip, up to a limit.
Limit	f_s,max = $\mu_s N$; f_s can be smaller.
Kinetic f_k	Opposes sliding; model size $\mu_k N$.
Normal N	Find the perpendicular force balance first.

B · KEY EQUATIONS + UNITS

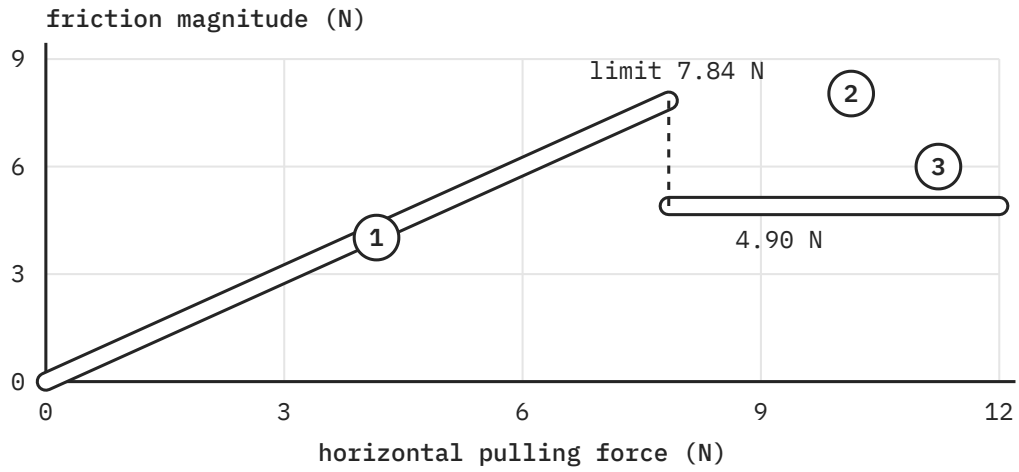
$$0 \le f_s \le \mu_s N$$

$$f_k = \mu_k N$$

μ no units
N mg here: level floor

C · COLOR IT · BEFORE AND AFTER SLIP

2 kg; μ_s = .40; μ_k = .25.



☐ **1 · ORANGE** Static friction

☐ **2 · ORANGE** Static limit: **7.84 N**

☐ **3 · ORANGE** Sliding level: **4.90 N**

D · WORKED EXAMPLE

For this box, compare a **6 N** pull while at rest with a **10 N** pull after sliding starts. Use g = **9.8 m/s²**; pulls are horizontal.

GIVEN	EQUATION	WORK	ANSWER
N = 2(9.8) = **19.6 N** μ_s = 0.40 μ_k = 0.25	Check $\mu_s N$. Sliding: use $\mu_k N$, then net force.	f_s,max = **7.84 N** at rest: f_s = **6 N** a = (10 − 4.9)/2	**0; 2.55 m/s²** At rest, then sliding.

E · THE 2026-XJ THREAD

DAY 17 OF 22 · NEW TOOL: FRICTION STATES

Surface friction is not a force that continually brakes XJ in empty space. Gravitational forces can still change its velocity, and atmospheric drag would matter in an atmosphere.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Friction changes when slipping begins

Adjustable static friction and the kinetic-friction model

If I push gently on a heavy box and it stays still, friction has matched my push. Pushing a little harder can make friction grow while the box remains at rest. That behavior shows why we cannot assign one fixed friction force to every situation. We first need to decide whether the surfaces are slipping relative to each other.

Static friction prevents relative slipping when it can. Its magnitude adjusts from zero up to a limit: $\mathbf{0 \le f_s \le \mu_s N}$. The coefficient μ is dimensionless, and the subscript s identifies the static case. The product of the coefficient and normal force gives the maximum available static friction, not necessarily the friction currently acting.

Our box has mass **2 kg** on a level surface, with a horizontal pull and no vertical acceleration. Therefore $\mathbf{N = mg}$ = **(2 kg)(9.8 m/s²)** = **19.6 N**. With $\mathbf{\mu_s}$ = **0.40**, the static limit is $\mathbf{f_{s,max}}$ = **(0.40)(19.6 N)** = **7.84 N**. We find the normal force first because another vertical force would change this limit.

Now pull horizontally with **6 N** while the box is at rest. Static friction can supply 6 N in the opposite direction because that is below its 7.84 N limit. The net force is **6 N – 6 N = 0 N**, so acceleration is zero. Using 7.84 N as the actual friction here would falsely predict a backward acceleration from a gentle forward push.

Once the box slides, we use the kinetic-friction model $\mathbf{f_k = \mu_k N}$. With $\mathbf{\mu_k}$ = **0.25**, its magnitude is $\mathbf{f_k}$ = **(0.25)(19.6 N)** = **4.90 N**. For a **10 N** pull in the direction of sliding, $\mathbf{F_{net}}$ = **10 N – 4.90 N** = **5.10 N**, giving $\mathbf{a}$ = **5.10 N ÷ 2 kg** = **2.55 m/s²** forward.

The graph shows static friction rising with the pull until the limit, followed by a lower kinetic-friction level once sliding occurs. That level is constant in our simplified model because the normal force and kinetic coefficient stay constant. Real surfaces can behave more complicatedly, but this distinction between adjusting static friction and modeled sliding friction is the starting point.

Friction opposes relative slipping, or the tendency to slip, at the contact. It does not always oppose an object's overall motion in every situation. A shoe pushing backward on the ground can receive forward static friction that helps a person walk. I want the contact and its slipping tendency identified before we choose the arrow's direction.

For our fictional asteroid, 2026-XJ, empty space does not provide a surface that continually applies sliding friction. Gravity can still change its velocity, and atmospheric drag would matter inside an atmosphere. Naming the actual interaction keeps our forecast from including a familiar force where its physical source is absent.